

MicroInspect

Bare-Board PCB Defect Detection

A comparative Bare-Board PCB inspection system that detects and classifies six manufacturing defects using classical topology and YOLOv11 deep learning.

Group 04

E/23/035 • E/23/104 • E/23/117 • E/23/336

Why bare-board PCB inspection is difficult

The challenge is not only classification, but also the defect must remain visible at the point of detection.

01 · Tiny defects

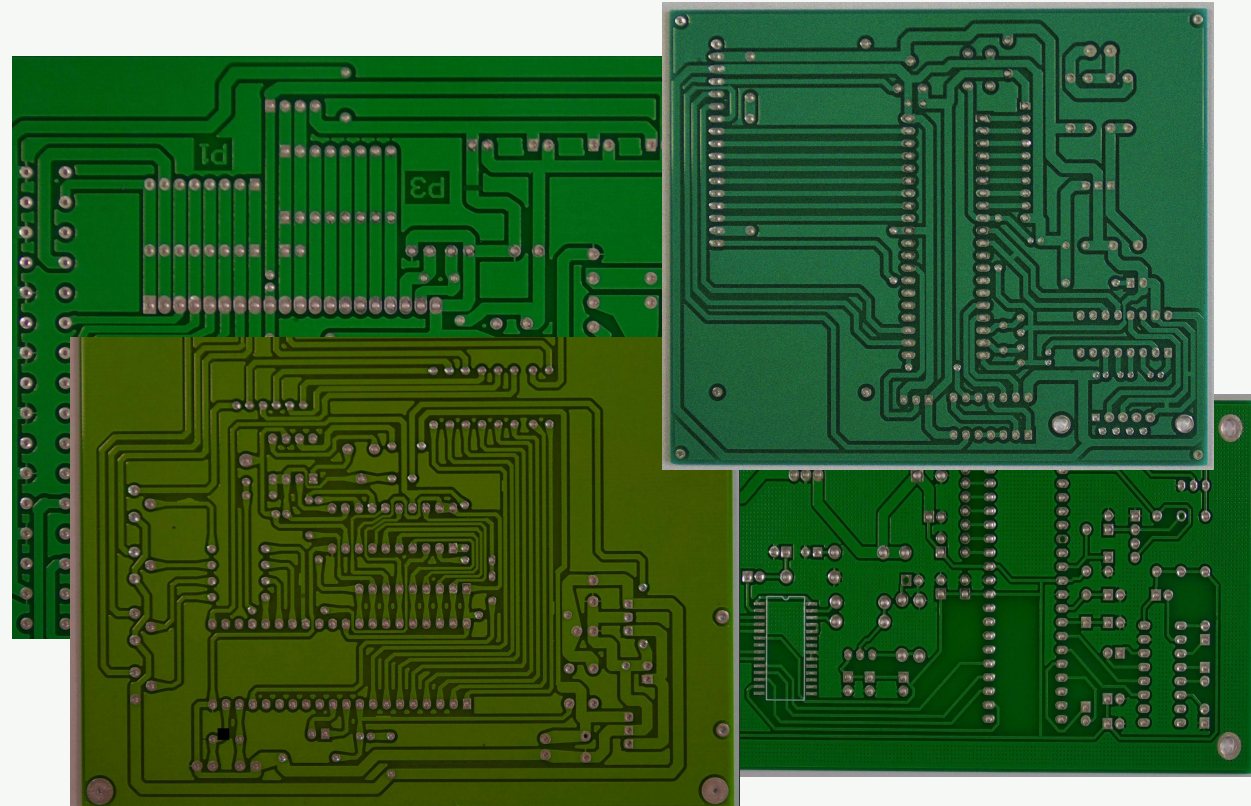
A full-board image contains defects that occupy only a small number of pixels.

02 · Similar appearance

Spur, Short and Spurious copper are all additive defects with subtle structural differences.

03 · Inspection reliability

Missing a real defect is more critical than misclassifying a detected defect.



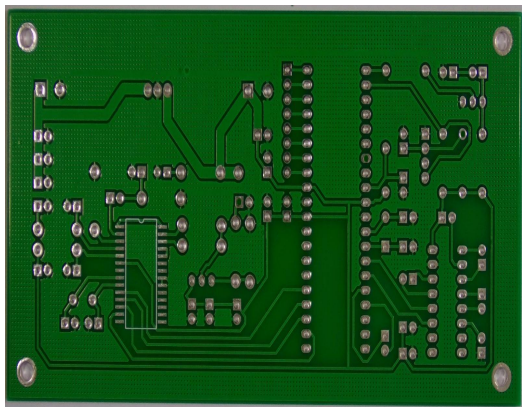
Printed Circuit boards contains defects at very different scales and positions.

Six target classes: Missing hole · Mouse_bite · Open_circuit · Short · Spur · Spurious_copper

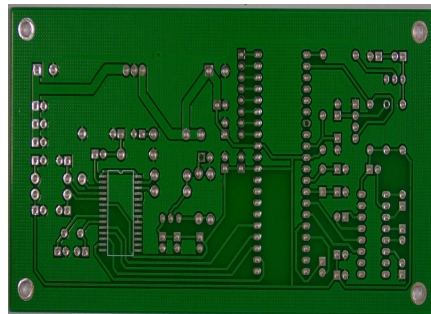
The visibility problem and why tiling was needed?

Directly resizing a high-resolution board compresses the very small detail we need to detect.

DIRECT RESIZE → DETAIL LOSS

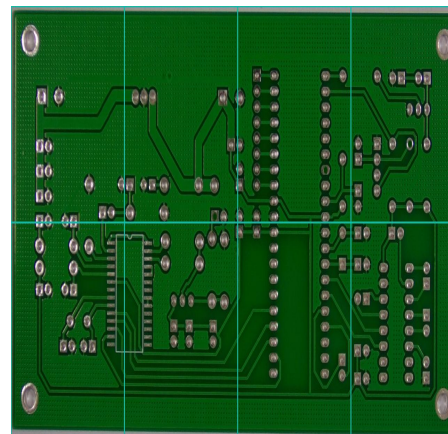


Whole board = 3034×1586

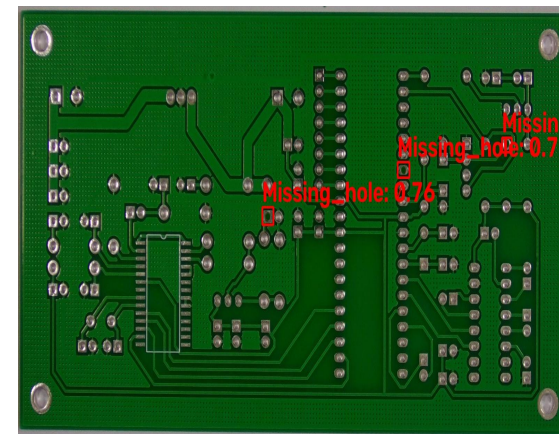


Whole board → 640×640
small defects shrink
further

OVERLAPPING TILES → PRESERVED DETAIL



640×640 tiles
15% overlap



Defect remains
locally visible

Dataset: We have six defect categories, and the labelled images are separated into training, validation, and test sets.

Sampling: keep 100% of defect-containing tiles; retain 10% of empty tiles to limit background dominance.

Two Approaches to Defect Detection

One method encodes PCB structure explicitly, the other learns visual features from data.

CLASSICAL TOPOLOGICAL PIPELINE

Golden PCB + Defect PCB

template image and test image

ORB + RANSAC

alignment

CLAHE + AbsDiff + Otsu

candidate mask

RETR_TREE + topology

structural reasoning

6 defect classes

decision

YOLOv11-MEDIUM PIPELINE

High-resolution board

input

640×640 overlapping tiles

local visibility

YOLOv11m

learned detection

Global coordinate remap

full-board boxes

Class-aware NMS

final detections

- Both methods find the defect and its location
- But they use different approaches: Explicit structure vs. learned representation.

Classical inspection: Detecting Defects Through PCB Structure

Instead of learning appearance, the pipeline aligns a golden board and uses topology to explain what changed.

METHODOLOGY

- 01 Align**
ORB + RANSAC registers the test image to the golden template.
- 02 Difference**
CLAHE → AbsDiff → Otsu produces a candidate defect mask.
- 03 Topology**
RETR_TREE finds hole / contour relationships; dilation tests local connectivity.
- 04 Classify**
Copper intersections distinguish Missing Hole, Mouse Bite, Open Circuit, Short, Spur and Spurious Copper.

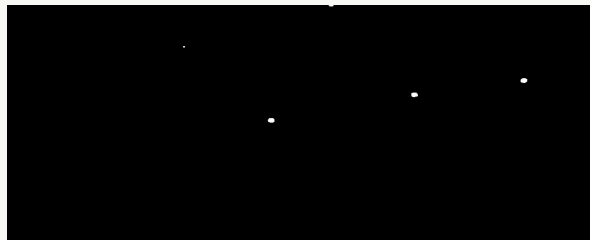
STRENGTHS

- Interpretable: every decision has a visible structural reason.
- Works without training a machine learning model.
- Uses PCB structure and trace connections to detect defects.
- Classifies all six defects using predefined rules.

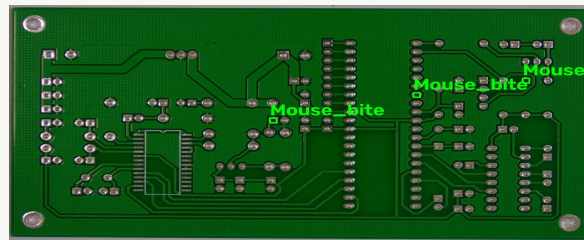
WEAKNESSES

- Depends on reliable template alignment; ORB alignment can fail on low-feature boards.
- Fixed thresholds are sensitive to board layout and scale.
- 11×11 Dilation can confuse Spur with Short when traces are close.
- Partial hole edges can cause Missing Hole detection to fail.

EVIDENCE FROM THE PIPELINE



Difference / candidate mask



Topology-based final detection

Key idea:
structure turns pixel change into a defect explanation.

PIPELINE EVOLUTION

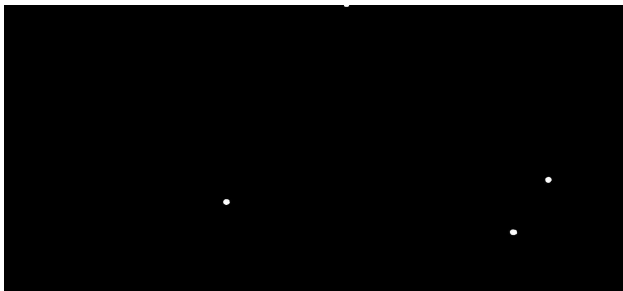
From pixel difference → structural reasoning → learned detection

Each stage addresses a limitation exposed by the previous one.

01 · PIXEL DIFFERENCE

Fast baseline

AbsDiff + threshold finds changed areas, but cannot reliably tell all six defect types apart.

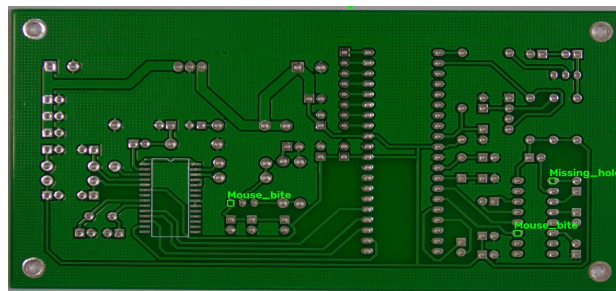


Example output: difference-based

02 · CLASSICAL TOPOLOGICAL

Interpretable improvement

Uses contour relationships (Which regions are inside or outside other regions) and trace connections to understand PCB structure.

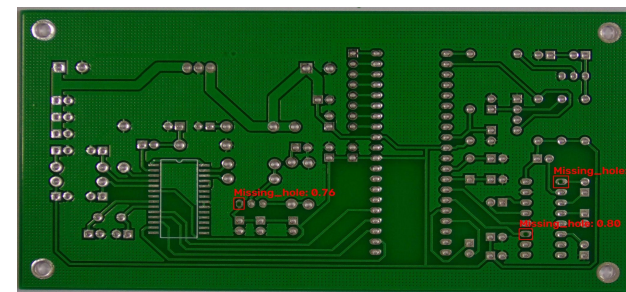


Example output: topology-based

03 · DEEP LEARNING

Learned detector

Learns visual + spatial features and avoids dependence on a golden template during detection.



Example output: YOLO detection

Our progression: From detecting changes → understanding structure → learning defects

QUANTITATIVE RESULTS

YOLOv11-Medium: stronger model, stronger localization

The YOLOv11-Medium model improved both overall detection and stricter localization compared with YOLOv11-Nano, which was our first experimentation.

mAP@0.50

92.01%

YOLOv11-Medium

- 100 epochs
- batch 8
- 640 x 640 px tiles

Precision

97.16%

Recall

90.97%

mAP @ 0.50:0.95

67.25%

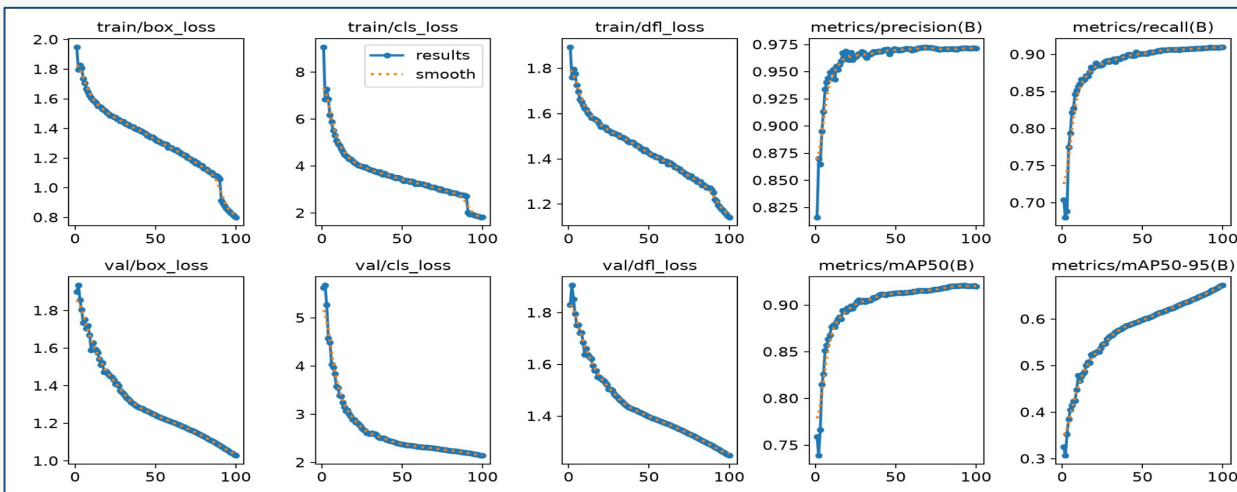
NANO → MEDIUM +1.40% mAP@0.50 +7.07% mAP@0.50:0.95 +1.64% Recall

- A detection is correct if $\text{IoU} \geq 0.50$ and the class is correct
- (mAP@0.50 / mAP@0.50:0.95 are YOLO AP metrics)

What this means

The higher mAP@0.50:0.95 shows better localization at stricter IoU levels. Some small or low-contrast defects are still missed.

YOLOv11-Medium training curves: decreasing losses and steadily improving detection metrics over 100 epochs.



VISUAL COMPARISON

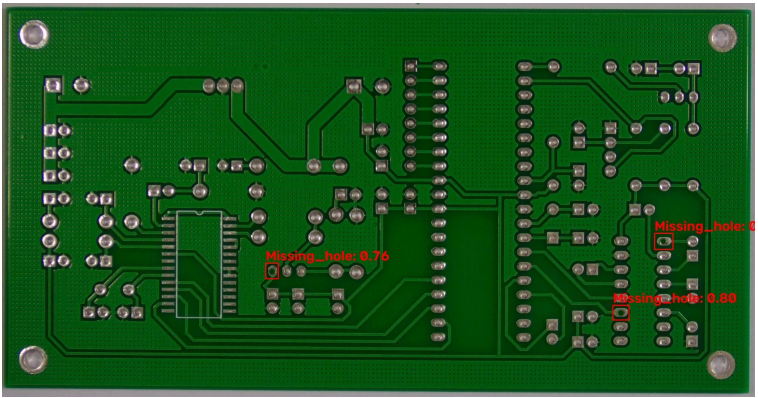
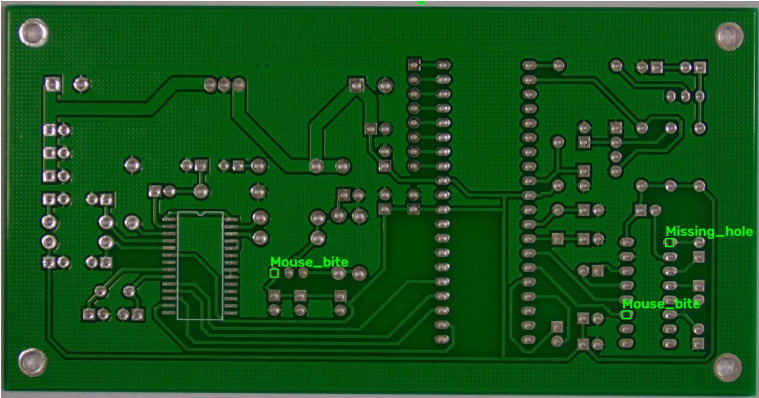
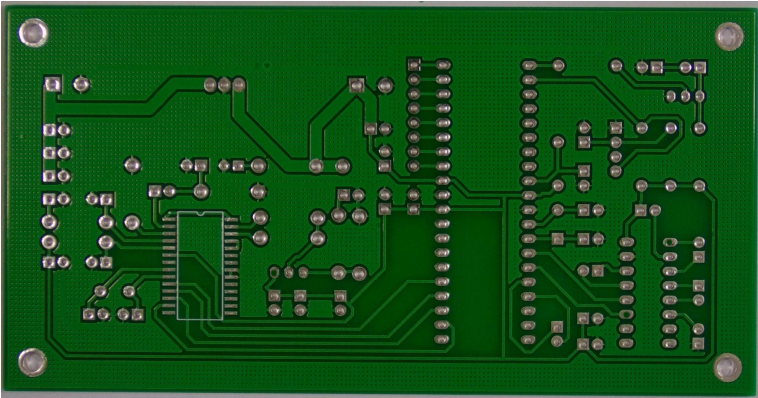
Same PCB, two inspection methods

Test Image

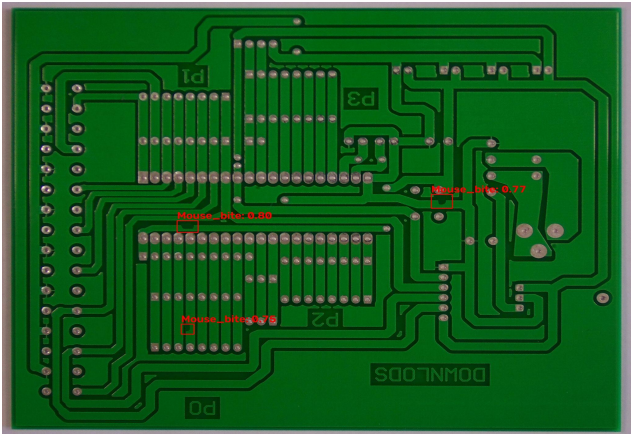
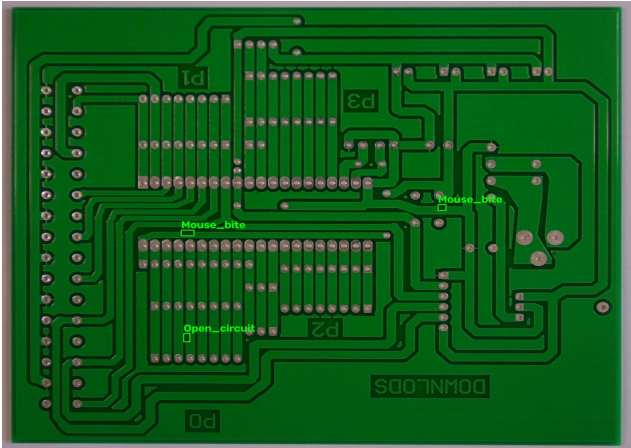
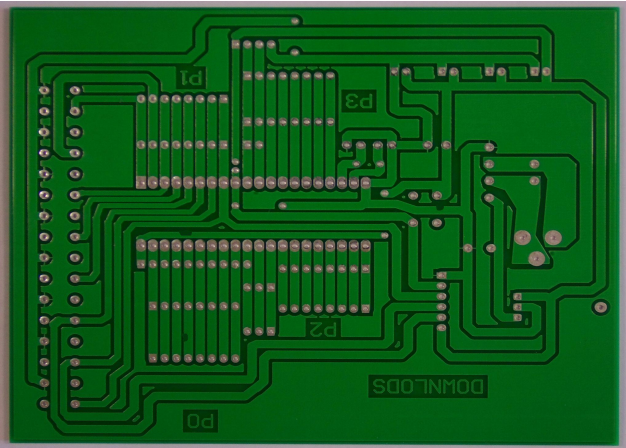
Classical Topological Result

Deep Learning YOLO Result

01. Missing Hole



02. Mouse Bite



VISUAL COMPARISON

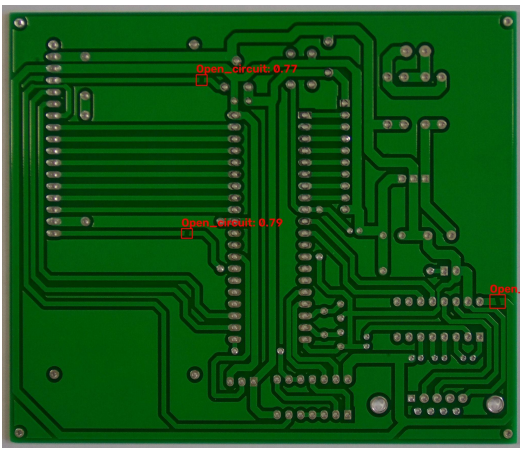
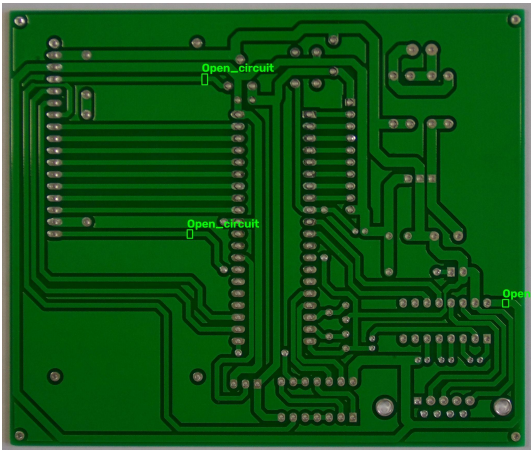
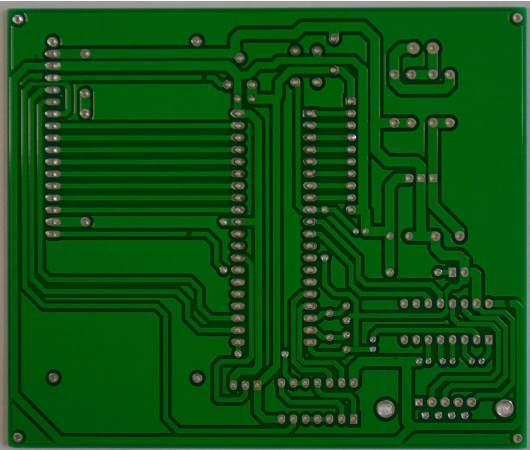
Same PCB, two inspection methods

Test Image

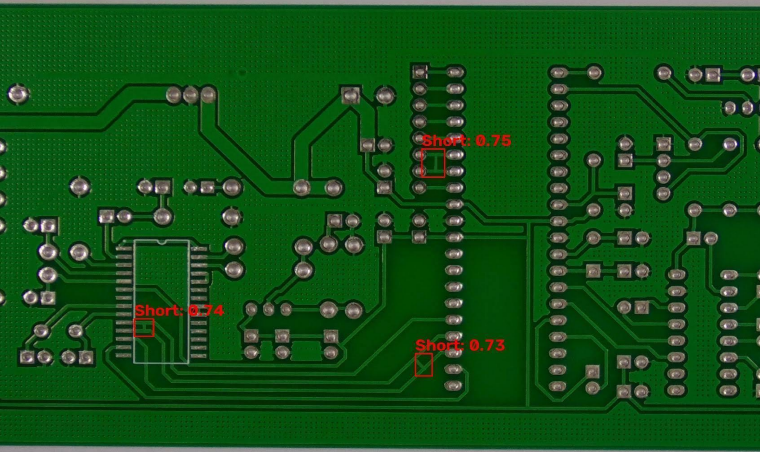
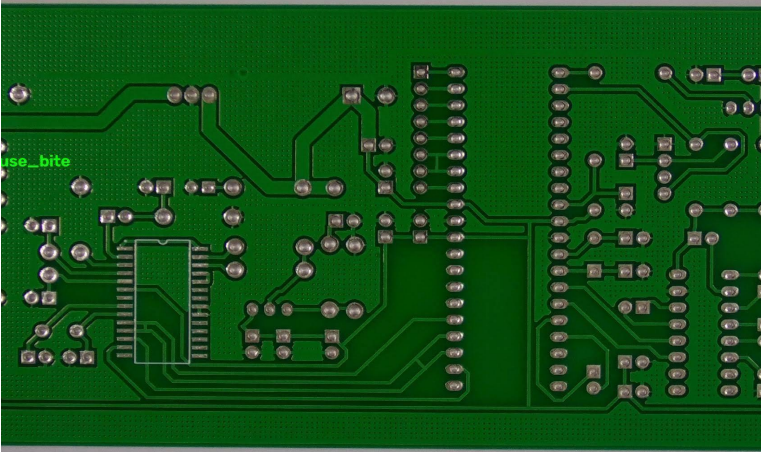
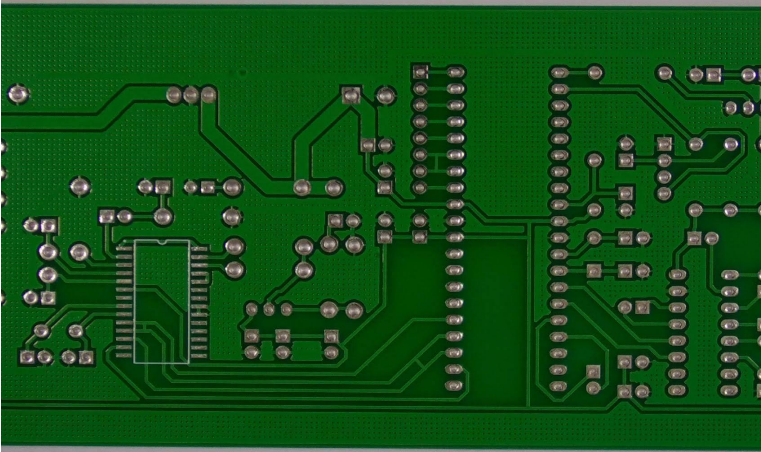
Classical Topological Result

Deep Learning YOLO Result

03. Open Circuit



04. Short Circuit



VISUAL COMPARISON

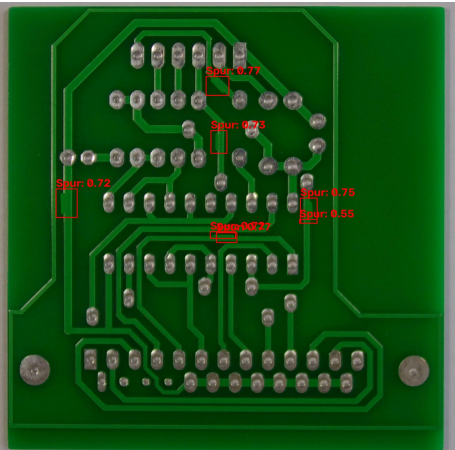
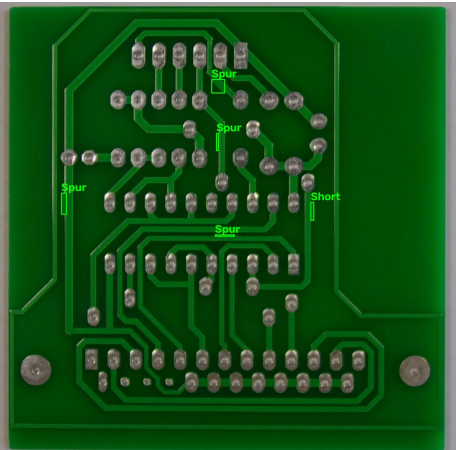
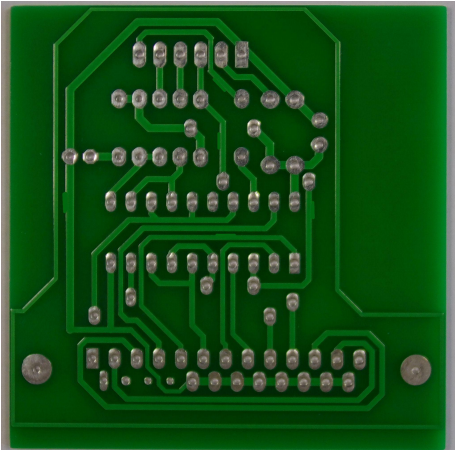
Same PCB, two inspection methods

Test Image

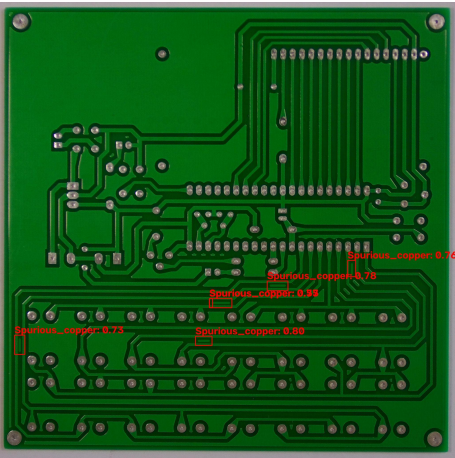
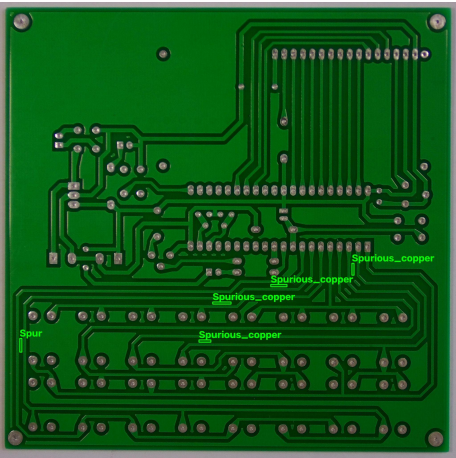
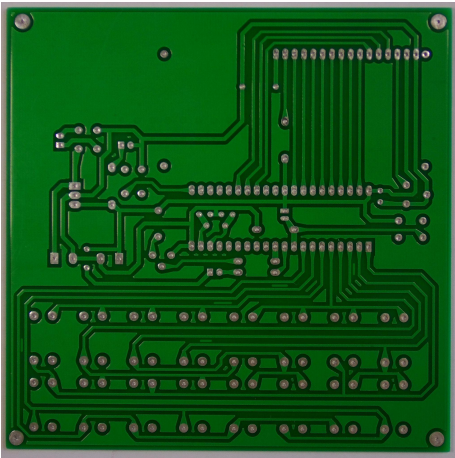
Classical Topological Result

Deep Learning YOLO Result

05. Spur

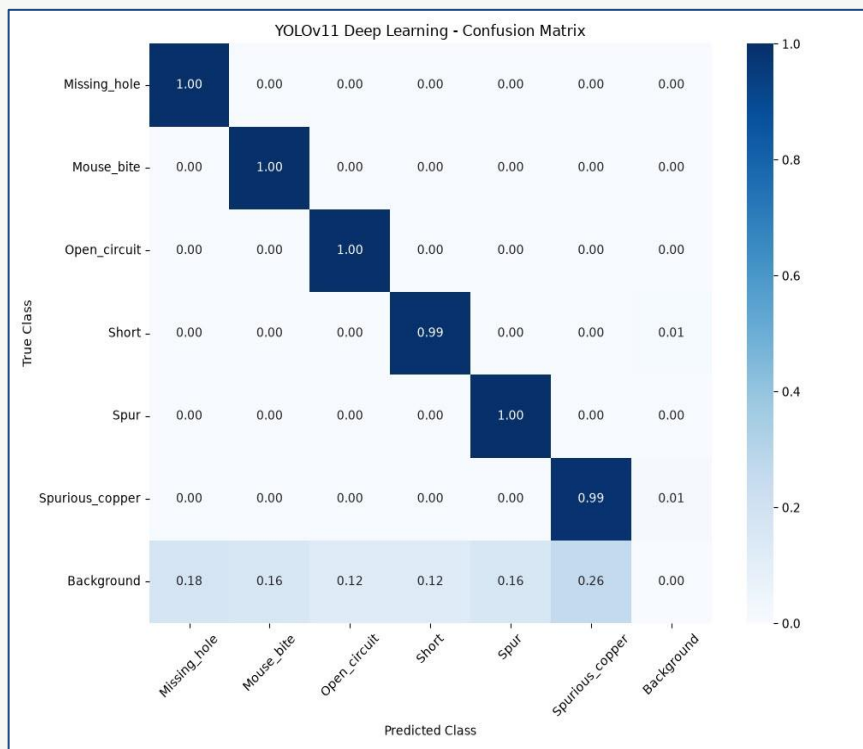


06. Spurious Copper Region



What mistakes still prevent perfect detection?

The key is understanding why each method fails, not just which one performs better.



The key failure is **Background errors** → **Defect** - Most errors come from background areas being detected as defects.

TOPOLOGICAL METHOD · WHERE IT BREAKS

- ORB alignment may fail when the board has few visual features.
- 11×11 dilation can turn a Spur touching another trace into Short.
- Partial hole edges can cause Missing Hole detection to fail.

YOLO DL METHOD · WHERE IT BREAKS

- Small or low-contrast defects may be missed.
- Spur and Short can look very similar.
- Tiling increases processing time for full-resolution boards.

Strong class separation; remaining errors are coming from background regions incorrectly detected as defects.

HARDEST PAIRS TO CLASSIFY

1) Spur vs. Short: similar looking defects

2) Mouse Bite vs. Open Circuit - partial vs. full break

MicroInspect: what we learned

Comparing both methods revealed their strengths and weaknesses.

WHAT WE DELIVERED

- Classical Topological method for all six defect classes.
- YOLOv11-Medium with high-resolution overlapping tiles.
- Maps detections back to the full PCB and removes duplicates using NMS
- FastAPI + browser user interface for testing and demonstrations of both implementations

THE ENGINEERING INSIGHT

- Detection accuracy alone is not enough for PCB inspection.
- The classical method provides structural understanding, while YOLO provides stronger learned detection.
- The comparison revealed complementary failure modes:
 - classical vision is sensitive to alignment and handcrafted rules
 - YOLO can miss subtle defects and produce false detections in complex background regions.

FUTURE POSSIBILITIES: HYBRID EXPLORATION

YOLO localization → topology-based refinement → final defect class

- **Find more defects:** Detect small and low-contrast defects.
- **Reduce false detections:** Check uncertain results using PCB structure.
- **Save processing time:** Use topology only for difficult cases.